



SHAH NEUROVISION
SPORTS TRAINING

A NEUROVISION EDGE GOALKEEPER GUIDE

SAVED BEFORE CONTACT

How goalkeepers read the penalty taker's run-up, plant foot, hips and trunk before the ball is struck, and a five-drill protocol to train it.

SEE → PROCESS → ACT



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FEATURING THE PENALTY CUE READ
EVIDENCE-REFERENCED EDITION

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HOW TO READ THIS GUIDE

THREE KINDS OF STATEMENT. ALWAYS LABELLED.

Sports vision content often blurs what was measured with what someone believes. This guide keeps them apart. Every section tells you which of the three you are reading.

RESEARCH SHOWS

A finding from peer-reviewed or published research, with a short citation. Full references and links are on pages 20 and 21.

NEUROVISION METHOD

A practical training recommendation built on that research. It is how we apply the evidence, not a finding in itself.

EMERGING AI

Early or experimental work in computer vision and machine learning. Promising, not yet a validated product for goalkeepers.

A NOTE ON DIRECTIONS

Every call in this guide (left, right, centre) is made from **your** point of view as the goalkeeper, facing the penalty taker. Research on reading height before contact is weaker than for left and right, so the drills focus on side first. [Higueras-Herbada et al., 2020](#)



THE 0.4-SECOND PROBLEM

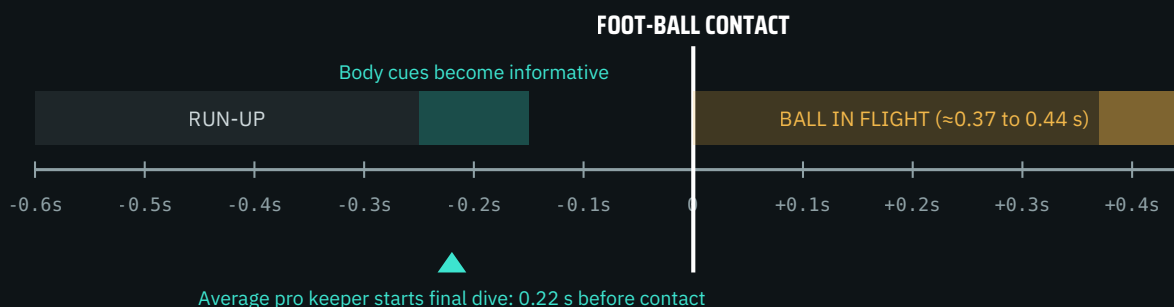
A goalkeeper who waits to see where the ball goes has usually already lost. The ball is simply too fast for a pure reaction.

11 m

Distance from the penalty mark to the goal line.

0.37-0.44s

Time for the ball to cover that distance at 30 m/s and 25 m/s. Illustrative arithmetic, not a measured average.



Timeline assembled from Hunter et al., 2018 (dive timing, 330 professional shootout penalties); Faquin et al., 2023 and Lopes et al., 2014 (when body cues carry direction information). Ball-flight band is illustrative arithmetic.

RESEARCH SHOWS

The ball reaches the goal in less time than it takes to start and complete the whole-body dive needed to reach a well-placed shot. [Diaz et al., 2012](#)

In 330 penalties from professional shootouts, goalkeepers started their final dive on average 0.22 s *before* the ball was struck. [Hunter et al., 2018](#)

That creates a trade-off. In a video study of more than 500 soccer-playing adults, direction calls stopped 0.4 s before contact were correct 55 to 64% of the time. Calls made at contact were correct roughly 80 to 90% of the time. [Hunter et al., 2018](#)

RESEARCH SHOWS

Expert and more successful keepers tended to **wait longer** before committing, and were more accurate for it. [Savelsbergh et al., 2002; 2005](#)

NEUROVISION METHOD

**READ EARLY.
COMMIT AS LATE AS YOU CAN
AFFORD.**

The skill is not guessing sooner. It is extracting reliable information sooner, so the late commitment you make is the right one.

WHERE EXPERT EYES GO

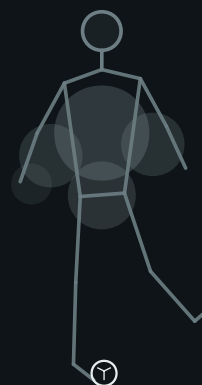
When researchers fitted goalkeepers with eye trackers, the better readers did not look at more things. They looked at fewer things, for longer, in more useful places.

EXPERT KEEPERS



Kicking leg, non-kicking (plant) leg and ball area, especially as contact approached.

NOVICE KEEPERS



Longer on the trunk, arms and hips, spread across more areas.

Schematic of reported gaze patterns, not a reproduction of study heat maps. [Savelsbergh et al., 2002](#)

RESEARCH SHOWS

- Expert keepers were more accurate, waited longer to respond, made fewer corrections, and used **fewer fixations of longer duration** on fewer areas of the scene. [Savelsbergh et al., 2002](#)
- Among experts, the more successful keepers spent longer fixating the **non-kicking (plant) leg** and predicted both height and direction better. [Savelsbergh et al., 2005](#)
- Across 42 sport studies, experts consistently used fewer, longer fixations and longer "quiet eye" periods than non-experts. [Mann et al., 2007](#)
- In live penalties, the final steady fixation (quiet eye) lasted longer on saves than on goals, and goals were associated with more gaze shifts between locations. [Piras & Vickers, 2011](#)

PLANT FOOT AND HIPS?

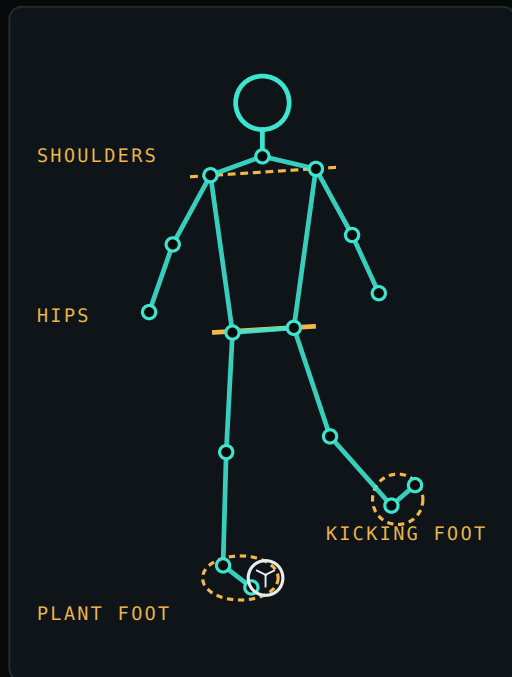
Eye-tracking points most clearly to the legs, especially the plant leg. The hips were where *novices* spent longer. Yet movement studies show the hips and trunk do carry direction information (next page). Use the hips as a check, not your anchor.

VIDEO IS NOT THE PITCH

When keepers had to actually attempt saves, they split their gaze more evenly between the taker and the ball than in video tasks. Video drills build the read. Live reps must follow. [Dicks et al., 2010a](#)

WHAT THE BODY GIVES AWAY

Motion-capture studies ask a different question from eye-tracking: not where keepers look, but which parts of the taker's movement actually relate to where the ball goes.



Keypoint figure at plant, the moment the support foot lands. The dots are the kind of body landmarks a pose-estimation system tracks (page 16).

NEUROVISION METHOD

Anchor your eyes low: plant foot first. Let hips, trunk and shoulders confirm or contradict it in your peripheral view.

RESEARCH SHOWS

PLANT FOOT

Where the support foot lands and which way it points. Support-foot orientation, its position relative to the ball and approach angle together predicted left versus right in one kinematic study. Several studies agree its orientation is informative about 200 to 250 ms before contact.

[Li et al., 2015](#); [Lopes et al., 2014](#)

HIPS

Pelvic orientation and rotation. Hip angle is among the body variables that vary with horizontal kick direction. Around plant, hip and torso orientation differ between shots going left and right.

[Lopes et al., 2014](#); [Hunter et al., 2018](#)

TRUNK AND SHOULDERS

Rotation of the torso toward the target side. In U19 players, trunk rotation was the main predictor of horizontal direction 250 and 150 ms before contact.

[Faquin et al., 2023](#)

KICKING LEG AND FOOT

Kicking-foot angle, swing direction and knee angle vary with direction. The foot's final path at contact is the most reliable cue of all, but it arrives last.

[Lopes et al., 2014](#); [Hunter et al., 2018](#)

RUN-UP

Approach angle contributed to predicting direction alongside the support foot. Treat the run-up as context: it sets expectations but has less direct evidence than the plant foot, hips and trunk.

[Li et al., 2015](#)

WHY ONE CUE IS NOT ENOUGH

The most useful information seems to live in how the movement unfolds across the body, not in a single tell.



RESEARCH SHOWS

DIRECTION IS SPREAD ACROSS THE BODY

Using motion capture and a new method of analysing whole-body movement, researchers found direction information both in single segments and distributed across several. Observers' judgments were best explained by the distributed information, with only a small contribution from any single local cue. [Diaz et al., 2012](#)

TAKERS CAN DISGUISE

When takers tried to fake the opposite side, some body variables were affected before contact but not at contact itself. [Lopes et al., 2014](#) In live trials with deceptive takers, early information from the run-up actually hurt keepers' performance. [Dicks et al., 2010b](#)

RESEARCH SHOWS

TECHNIQUE CHANGES THE READ

Fast shots with the side of the foot revealed direction earlier. Slower side-foot shots gave away much less in the early part of the kick. [Hunter et al., 2018](#)

HEIGHT COMES LATER

Height is predicted less clearly from the taker's body than side. Keepers began the sideways part of the save before contact but the vertical part about 245 ms after, using ball flight. [Higuera-Herbada et al., 2020](#) Kicking-foot height predicted vertical direction only at contact. [Faquin et al., 2023](#)

NEUROVISION METHOD

Agreement beats any single tell. When plant foot, hips and trunk point the same way, your read is stronger. When they disagree, treat it as a warning sign of disguise or a late shot, and hold your commitment a little longer.

SEE → PROCESS → ACT

Every penalty is the same three-stage loop. Train each stage on purpose, then join them.

SEE

FIND THE RIGHT
INFORMATION

- Settle your gaze low on the plant-foot and ball area as the run-up starts.
- Use few, long fixations. Resist darting between face, arms and ball.
- Keep hips and shoulders in peripheral view.

Savelsbergh et al., 2002; 2005;
Mann et al., 2007

PROCESS

TURN CUES INTO A READ

- Plant foot: which way does it point on landing?
- Do hips and trunk agree with it?
- Agreement means a stronger read. Conflict means caution.
- Form a side call before contact.

NEUROVISION METHOD

ACT

COMMIT DECISIVELY

- Commit to the side call as late as your dive allows.
- Decide height from ball flight, not the body.
- Go fully. A half-dive loses both ways.

Savelsbergh et al., 2005;
Higueras-Herbada et al., 2020

NEUROVISION METHOD

BRIDGE FROM SCREEN TO PITCH

Drills 1 to 5 train SEE and PROCESS on video. Because gaze behaves differently when you have to make a real save [Dicks et al., 2010a](#), finish each week with live reps: a teammate takes penalties, you call the side out loud as the plant foot lands, then dive. Log the call, not just the save.

KEY IDEA

Penalties are not saved by reflexes. They are saved by what you saw a moment before anyone else was ready to look.



BEFORE CONTACT, CHECK:

01 RUN-UP

What direction is the taker approaching from?

Angle and straightness of the approach. Context, not a verdict.

02 PLANT FOOT

Where is it placed relative to the ball?

Which way do the toes point as it lands? Your anchor cue.

03 HIPS

Are the hips opening or closing?

Pelvis turning toward one side as the plant foot lands.

04 ANKLE / FOOT

What direction is the kicking foot oriented?

Foot angle and swing path. Reliable but arrives last.

05 TRUNK

Where is the torso rotating?

Torso rotation in the last 250 ms before contact.

06 SEQUENCE

Do several cues point the same way?

Agreement: commit. Conflict: hold a beat longer.

EYES LOW. FEW LOOKS. LONG LOOKS.

Plant foot and ball area first. Hips and shoulders in peripheral view.

SIDE FROM THE BODY. HEIGHT FROM THE BALL.

Commit to the side as late as you can afford.

Evidence behind each cue: pages 4 to 6. Cues are tendencies, not guarantees. Takers can disguise, and technique changes how early direction shows.



THE PENALTY CUE READ

Five drills, from simple to advanced. Each one isolates part of the SEE → PROCESS → ACT loop, then Drill 5 turns it into a daily habit you can do without a ball.

1

FREEZE-FRAME READ

Early information → prediction → decision

EASY

2

PROGRESSIVE OCCLUSION

How early can you make a useful read?

BUILDS

3

CUE ISOLATION

Which cues carry your read?

TARGETED

4

CONFIDENCE CALL

Is your confidence calibrated?

ADVANCED

5

NO-BALL TRAINING

Daily reps with instant feedback

ANYWHERE

RESEARCH SHOWS

Skilled goalkeepers who trained at home with temporally occluded penalty videos **and feedback** improved their prediction accuracy over eight weeks. A group that only watched televised shootouts did not. [Murgia et al., 2014](#)
Perceptual training with either explicit cue instruction or a more implicit approach improved prediction of penalty direction. [Faquin et al., 2024](#)

NEUROVISION METHOD

- **Dose:** 15 minutes, 3 to 4 times a week.
- **Clips:** 20 per session. Always get the answer straight after each call.
- **Baseline, then retest** at week 6 on clips you have never seen (page 15).
- **Stopping rule:** end the session when attention drops. Tired calls teach guessing.

WHAT YOU NEED

A phone, tablet or laptop; penalty footage filmed from behind the goal (keeper's-eye view) at 30 or 60 frames per second; a video player that steps frame by frame; and the score log on page 19. Best source: film your own team's takers. It is legal, representative, and the answer is known.



1 FREEZE-FRAME READ

TRAINS: SEE, THEN DECIDE

NEUROVISION METHOD

HOW TO RUN IT

- 1 Load a penalty clip and pause it **just before** foot-ball contact.
- 2 Hide anything that shows the outcome: cover the lower screen or crop out ball flight and the keeper's dive.
- 3 Scrub back and study the run-up at normal speed, then frame by frame.
- 4 Lock onto the plant foot, then check hips, ankles and body position.
- 5 Call it out loud: **LEFT**, **RIGHT** or **CENTRE** (your view).
- 6 Unfreeze the clip and watch the outcome.
- 7 Record the call and the result on the score log.

DOSE

20 clips. No time pressure in this drill: take as long as you need. Speed comes later.

WHAT THIS TRAINS

EARLY VISUAL INFORMATION

→ PREDICTION

→ DECISION

You are building a vocabulary: what a left-side plant foot, hip opening and trunk turn look like, repeated until you recognise them.

RESEARCH SHOWS

Keepers can train prediction with video clips stopped before contact, provided each call gets feedback. [Murgia et al., 2014](#)

COACH CUE

"Where did the foot land, and where is it pointing?" Ask it before every call.

SCORING

Correct = 1. Incorrect = 0. With three choices, pure guessing lands near 33%. Your first job is to beat your own baseline, not someone else's score.

2

DRILL 2 OF 5 · THE PENALTY CUE READ

PROGRESSIVE OCCLUSION

TRAINS: READING EARLIER

NEUROVISION METHOD

HOW TO RUN IT

- 1 Find the exact contact frame in each clip (T0).
- 2 Make versions that cut to black at the points below. Count frames back from T0.
- 3 Show clips once at full speed. No replays, no scrubbing.
- 4 Call within two seconds of the cut. Log it.
- 5 Start at T-100. Move one step earlier only when your accuracy at the current step is steady across two sessions.

CUT (MS)	FRAMES @30	FRAMES @60	NOTES
T-100	3	6	Start here. Plant foot and hips visible.
T-200	6	12	Within the range where pro keepers begin to dive.
T-300	9	18	Harder. Plant foot may still be landing.
T-400	12	24	Earliest point in the Hunter et al. study.
T-500	15	30	Optional stretch. Beyond the tested range.

These cut points are a **training framework**, not a universal standard. One frame is about 33 ms at 30 fps and 17 ms at 60 fps.

THE QUESTION IT ANSWERS

HOW EARLY CAN YOU MAKE
A USEFUL PREDICTION?

RESEARCH SHOWS

Temporal occlusion, cutting video at set points before contact, is the standard lab method for anticipation. One large study cut clips at 0.4, 0.3, 0.2 and 0.1 s before contact and at contact. Accuracy rose as the cut moved closer to contact. [Hunter et al., 2018](#)

The same study found professional keepers began their final dive about 0.22 s before contact on average, which is why the 100 to 400 ms range is the one that matters.

YOUR "EARLIEST USEFUL READ"

The earliest cut point where your accuracy stays clearly above chance (about 33% with three choices, 50% with two) across 20 or more clips. Track it over weeks. It is your number, and it should move earlier.



3

DRILL 3 OF 5 · THE PENALTY CUE READ

CUE ISOLATION

TRAINS: WHICH CUES CARRY YOUR READ

NEUROVISION METHOD

HOW TO RUN IT

Use your T-100 or T-200 clips. Run five rounds of 8 clips. In each round, you may base the call **only** on the named cue. Log accuracy per round.

ROUND 1

PLANT FOOT

Where did it land relative to the ball? Which way does it point?

ROUND 2

HIPS

Opening or closing? Which way is the pelvis turning as the foot lands?

ROUND 3

ANKLES / KICKING LEG

Foot angle and swing path of the kicking leg.

ROUND 4

TRUNK AND SHOULDERS

Which way is the torso rotating? Where are the shoulders aligned?

ROUND 5

FULL-BODY PATTERN

All cues together. Do they agree? Call it.

READ YOUR RESULTS

Your strongest round shows the cue you already use. Your weakest shows where to spend time. Many keepers find the full-body round beats any single cue, which is exactly what the research predicts.

WHAT THIS TRAINS

**FROM ISOLATED CUES →
INTEGRATED PATTERN**

RESEARCH SHOWS

Observers' direction judgments were best explained by information distributed across several body segments, not one local cue. [Diaz et al., 2012](#)

Directing attention to useful cues through perceptual learning has been used to improve penalty anticipation [Savelsbergh et al., 2010](#), and both explicit and implicit perceptual training improved prediction. [Faquin et al., 2024](#)

PROGRESSION

Weeks 1 to 2: all five rounds. Weeks 3 onward: only your two weakest cues plus Round 5. The destination is always Round 5. Isolated cues are scaffolding, and you should not stay on them.

WHY NOT JUST MEMORISE ONE TELL?

Takers can manipulate single cues before contact. [Lopes et al., 2014](#) A pattern is much harder to fake than one body part.



4

CONFIDENCE CALL

TRAINS: CALIBRATED COMMITMENT

NEUROVISION METHOD

HOW TO RUN IT

- 1 Use any occlusion clips from Drill 2.
- 2 Make each call as **direction + confidence**, out loud.
Example: “**Right, 70%.**”
- 3 Use steps of 10: from 40% (barely better than a guess) to 100% (certain).
- 4 Reveal the answer. Log direction, confidence and result.
- 5 After 40 or more calls, group them by confidence and work out your real hit rate in each group.

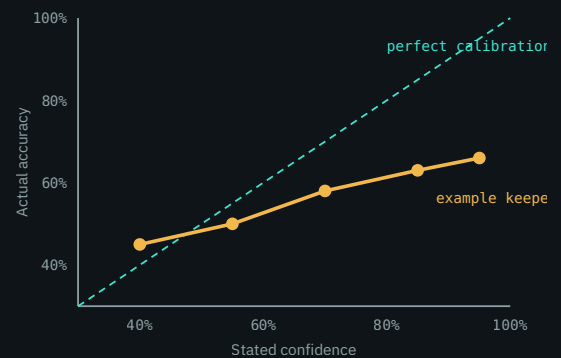
HOW TO READ IT

- **Overconfident:** your 80% calls land 55% of the time. You are committing on weak reads.
- **Underconfident:** your 50% calls land 75%. You see more than you trust. Commit.
- **Calibrated:** stated confidence roughly equals actual accuracy.

WHAT THIS TRAINS

DECISION QUALITY, NOT JUST ACCURACY

EXAMPLE CALIBRATION CHART



Illustrative example only. Not data from a real athlete or study.

IMPORTANT

Confidence is not proof a call was right. A certain call can be wrong, and a doubtful one can be right. This drill monitors how well your confidence tracks reality, so you can learn when a read is strong enough to act on.

5 NO-BALL TRAINING

TRAINS: DAILY REPS WITH FEEDBACK

NEUROVISION METHOD

HOW TO RUN IT

- 1 Build a playlist of 20 occluded penalty clips, mixed left, centre and right, from several takers.
- 2 Watch each clip once. Call the direction before the screen goes black or within two seconds.
- 3 Get the answer **immediately**: watch the uncut clip or check your answer list.
- 4 Score it on the log. Move to the next clip.
- 5 Once a week, swap in fresh clips so you are reading bodies, not memorising videos.

MEASURE	HOW TO SCORE
Direction	Correct = 1 point. Incorrect = 0.
Confidence (optional)	1 = guess · 2 = lean · 3 = fairly sure · 4 = sure · 5 = certain
Session score	Correct ÷ total clips × 100
Cut point	Log which occlusion point you used (T-100 to T-400)
Trend	Compare against <i>your</i> baseline, at the same cut point

There is no published benchmark score for this drill. Clips, takers and cut points vary too much for one number to mean anything. Your progress against your own baseline is what counts.

WHY NO BALL?

Reading is a visual and decision skill. You can load it with dozens of reps a day, anywhere, without a taker, a goal, or wear on your body. Then you take it onto the pitch.

RESEARCH SHOWS

In a randomised study, 42 skilled goalkeepers were split into three groups. Only the group that trained at home with occluded penalty videos **plus right and wrong feedback** improved their prediction accuracy after eight weeks. Watching TV shootouts did not help. [Murgia et al., 2014](#)

The lesson: feedback is the active ingredient. Watching is not training.

SESSION TEMPLATE · 15 MIN

0–2	Warm-up: 5 easy clips at T-100
2–12	Working block: 20 clips at your current cut point
12–15	Review misses frame by frame. What did you miss?



THE SIX-WEEK BLOCK

At the professional level we don't guess this. We test it. Baseline first, a planned training block, then a retest on clips you have never seen.

NEUROVISION METHOD

Wk 0	BASELINE	40 fresh clips: 20 at T-100, 20 at T-300. No feedback until the end. Record scores.
Wk 1-2	BUILD VOCABULARY	Drill 1 × 2 sessions, Drill 3 (all rounds) × 1, Drill 5 daily if you can.
Wk 3-4	READ EARLIER	Drill 2 progression × 2, Drill 3 (weak cues + full body) × 1, Drill 5.
Wk 5	COMMIT WELL	Drill 4 × 2, Drill 2 at your earliest useful read × 1, live reps × 1.
Wk 6	RETEST	Same format as baseline, on 40 clips you have never seen. Compare.

BUILD A CLEAN CLIP LIBRARY

- Film from behind the goal at keeper eye height, 60 fps if possible.
- Use many different takers, both footedness, both techniques.
- Split clips into a **training set** and a locked **test set**. Never train on test clips.
- Label each clip's true side before you cut it.

STAFF AND COACHES

- Run the baseline in preseason. Retest at week 6.
- Put Drill 5 in the warm-up window on penalty-heavy weeks.
- Track the earliest useful read and calibration alongside save percentage. Don't judge the block on a handful of match penalties.

Why a locked test set? If you retest on clips you trained on, a better score may just mean you remember those videos. Fresh clips measure the skill you actually care about: reading a body you have not seen before.



AI-ASSISTED PENALTY READING

Computer vision can already estimate body position from ordinary video. That makes it possible to measure, frame by frame and consistently, the same cues this guide trains you to see.

EMERGING AI

A conceptual system. No validated commercial product is implied.

01

VIDEO IN

Penalty clip from behind the goal

02

DETECT

Find the taker and the ball in each frame

03

POSE

Track body keypoints: feet, knees, hips, shoulders

04

ALIGN

Locate the contact frame (T0) and count back

05

MEASURE

Angles, rotations and timing for each cue

06

REVIEW

Frame-by-frame overlays, rep comparison, auto-cut drill clips

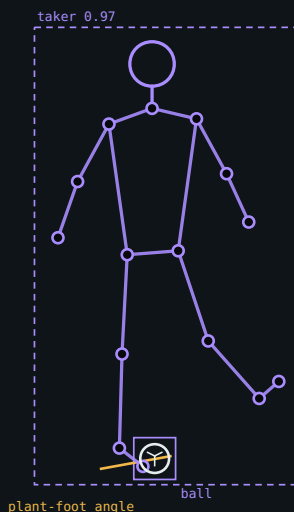


Illustration of what pose estimation outputs: a box around the taker, keypoints on the body, and measurements derived from them.

WHAT RESEARCH HAS ALREADY TRIED

EMERGING AI

A recent study built a dataset of 755 penalties from real match video and trained a model on body keypoints plus video frames from just before contact. It predicted left, centre or right with 89% accuracy on held-out clips, beating keypoint-only and video-only versions. [Ranasinghe & Ranasinghe, 2026](#)

Another group combined video segmentation with 3D pose estimation to predict which of eight goal zones a kick would reach, and compared which body parts mattered most. [Mauricio Salazar & Alatrística-Salas, 2024](#)

RESEARCH SHOWS

Lab studies already do a version of this with motion-capture suits, linking plant foot, hips and trunk to kick direction. [Diaz et al., 2012](#); [Lopes et al., 2014](#); [Faquin et al., 2023](#)

READ THIS CAREFULLY

These are early research results on specific datasets, mostly using movement very close to contact. They have not been shown to improve a goalkeeper's saves. No system predicts penalty direction perfectly, and takers can disguise.

SIX THINGS A SYSTEM COULD ANALYSE

EMERGING AI

Concept design based on the cues research has linked to kick direction.

MODULE	WHAT COMPUTER VISION COULD MEASURE	HOW A GOALKEEPER WOULD USE IT
RUN-UP	Approach angle; number and rhythm of steps; speed changes; final approach direction; body alignment in the last steps.	Compare a taker's run-ups across reps. Spot pauses and rhythm changes that come before disguise.
PLANT FOOT	Location relative to the ball; distance from it; foot angle and the direction it points; how that relates to where the ball went.	Freeze on the landing frame with an angle overlay. Train your eye to judge foot direction faster.
HIPS	Pelvic orientation and rotation; opening versus closing; rotation direction just before contact.	See hip rotation as a line across the pelvis. Check whether it agrees with the foot.
ANKLES AND LOWER LEG	Support-leg position; kicking-leg swing path; ankle and foot orientation before contact.	Review the swing in slow motion. Learn what side-foot versus instep preparation looks like.
TRUNK AND SHOULDERS	Torso rotation; shoulder alignment; lean; whole-body orientation.	Overlay the shoulder line. Notice rotation in the final 250 ms.
PRE-CONTACT SEQUENCE	Timing of each event relative to contact; whether cues agree or conflict.	See the whole chain on one timeline. Learn which combinations mean commit and which mean wait.

THE SEQUENCE IS THE POINT

RUN-UP → PLANT FOOT → HIPS → TRUNK → KICKING LEG → CONTACT

Research suggests direction information is spread across the body [Diaz et al., 2012](#). A system that lines up every event against the contact frame lets you review the whole chain, not one isolated tell.

WHERE AI FITS: TRAINING

- Auto-cutting occlusion clips at exact frames for Drill 2.
- Highlighting the cue for each round of Drill 3.
- Comparing a keeper's calls with the measured cues to show what was missed.
- Building scouting reels of one taker's patterns.

WHERE IT DOES NOT FIT

- Replacing the keeper's own read. The skill still has to live in your eyes.
- Promising a direction. Even the best research model makes errors.
- Live in-match prompting. This guide describes review and training only.



WHAT THIS GUIDE DOES NOT CLAIM

Good training is honest about its limits. Here are ours.

THIS WILL NOT GUARANTEE SAVES.

No study has shown that this protocol raises match save percentage. Research shows trained keepers predict better on video tests. That is a different, narrower claim.

[Murgia et al., 2014](#)

VIDEO SKILL DOES NOT FULLY EQUAL PITCH SKILL.

Gaze changes when you actually have to make the save, which is why live reps are built into the week.

[Dicks et al., 2010a](#)

TAKERS CAN DECEIVE.

Some body cues can be manipulated before contact, and early information can mislead keepers against deceptive takers.

[Lopes et al., 2014](#); [Dicks et al., 2010b](#)

SOME TAKERS DO NOT GIVE MUCH AWAY.

Slower side-foot shots reveal less early. Some takers also wait for the keeper to move first, which changes the whole duel.

[Hunter et al., 2018](#)

HEIGHT IS HARDER TO READ THAN SIDE.

Plan to read side from the body and height from the ball.

[Higueras-Herbada et al., 2020](#)

AI IS NOT A CRYSTAL BALL.

Research models are promising but tested on limited datasets. None predicts perfectly, and none has been shown to improve keepers' saves yet.

[Ranasinghe & Ranasinghe, 2026](#)

CUE TIMING VARIES.

The 100 to 400 ms cut points are a training framework drawn from the research, not a standard every taker follows.

[Hunter et al., 2018](#)

What the evidence does support: expert keepers read the body differently, the taker's body carries real direction information before contact, and goalkeepers can improve their video-based prediction with structured training and feedback. That is what this guide is built on.

SCORE LOG

THE PENALTY CUE READ

Name	Date	Drill #	Cut point (T-)	Week (0 to 6)
------	------	---------	----------------	---------------

#	TAKER / CLIP	CUE FOCUS	MY CALL L / C / R	CONFIDENCE 1-5	ACTUAL L / C / R	CORRECT 1 / 0
01						
02						
03						
04						
05						
06						
07						
08						
09						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						

Total correct: _____ / _____ =
_____ %

Baseline at this cut point: _____
%

Most-missed cue: _____

RESEARCH & REFERENCES

Links open the publisher or DOI record. Some articles sit behind a paywall; abstracts are free. Citations in the guide use author and year.

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ABOUT THE NUMBERS IN THIS GUIDE

Percentages and timings are quoted only where a cited study reports them. The ball-flight window on page 3 is simple arithmetic (11 m at 25 to 30 m/s), labelled as illustrative. The calibration chart on page 13 is an invented example. Drill doses, cut-point progressions and the six-week block are NeuroVision method, not research findings.



SHAH NEUROVISION
SPORTS TRAINING

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Dr. Shah is an optometrist with a fellowship in vision development (FOVDRA). Through Shah NeuroVision Sports Training and the NeuroVision Edge approach, he works on the visual side of sport: what athletes see, how they process it, and how they act on it.

This guide exists because the most important moment in a penalty happens before the ball moves. That moment can be trained.

SEE → PROCESS → ACT

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